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Detection of data storage device removal

Abstract

Detecting the removal of a data storage device from a storage system involves first determining that a shorter pin of an electrical connector of a storage device is disconnected from a mating electrical connector, such as by sensing a voltage drop on that pin, then determining at a later time that a longer pin of the connector is disconnected from the mating connector. Responsive to determining that the longer pin was disconnected after a predetermined period of time after the shorter pin, a conclusion may be made that the storage device has been removed from the system as opposed to being subject to a simple device power aberration. Thus, responsive data destruction action(s) may be taken to render the data stored on the device inaccessible to the attacker thereby protecting the device even after the device is removed from the storage system.

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Background/Summary

FIELD OF EMBODIMENTS
(1) Embodiments of the invention may relate generally to data storage systems, and particularly to approaches to detecting that a data storage device is removed from a data storage system.
BACKGROUND
(2) As networked computing systems grow in numbers and capability, there is a need for more storage system capacity. Cloud computing and large-scale data processing further increase the need for digital data storage systems that are capable of transferring and holding significant amounts of data. One approach to providing sufficient data storage in data centers is the use of arrays of data storage devices typically configured and provisioned as one or more data storage systems. Many data storage devices can be housed in an electronics enclosure (sometimes referred to as a “rack”), which is typically a modular unit that can hold and operate independent data storage devices in an array, as well as computer processors, routers and other electronic equipment. Data centers typically include many rack-mountable data storage devices mounted within respective slots of an enclosure and which are used to store the massive amounts of data.
(3) Data storage devices employ different measures of security to the storage system to avoid malicious operations. In flash storage devices, it is common practice to protect the data with firmware authentication and/or drive encryption. Physical removal of a storage device in some settings, such as a storage server, is highly irregular and may be the outcome of a malicious action.
(4) Any approaches that may be described in this section are approaches that could be pursued, but not necessarily approaches that have been previously conceived or pursued. Therefore, unless otherwise indicated, it should not be assumed that any of the approaches described in this section qualify as prior art merely by virtue of their inclusion in this section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings and in which like reference numerals refer to similar elements and in which:
(2) FIG. **1**A is a block diagram illustrating a solid-state drive (SSD), according to an embodiment;

(3) FIG. **1**B is a plan view illustrating a hard disk drive (HDD), according to an embodiment;
(4) FIG. **2** is a perspective view illustrating a data storage system, according to an embodiment;
(5) FIG. **3** is a block diagram illustrating a data storage system architecture, according to an embodiment;
(6) FIG. **4** is a diagram illustrating a system for device removal detection, according to an embodiment; and
(7) FIG. **5** is a flow diagram illustrating a method for detecting removal of a storage device, according to an embodiment.

DETAILED DESCRIPTION

(8) Approaches to detecting the removal of a data storage device from a data storage system are described. In the following description, for the purposes of explanation, numerous specific details are set forth to provide a thorough understanding of the embodiments of the invention described herein. It will be apparent, however, that the embodiments of the invention described herein may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form to avoid unnecessarily obscuring the embodiments of the invention described herein.

INTRODUCTION

Terminology

(9) References herein to “an embodiment”, “one embodiment”, and the like, are intended to mean that the particular feature, structure, or characteristic being described is included in at least one embodiment of the invention. However, instances of such phrases do not necessarily all refer to the same embodiment or to every embodiment.

(10) If used herein, the term “substantially” will be understood to describe a feature that is largely or nearly structured, configured, dimensioned, etc., but with which manufacturing tolerances and the like may in practice result in a situation in which the structure, configuration, dimension, etc. is not always or necessarily precisely as stated. For example, describing a structure as “substantially vertical” would assign that term its plain meaning, such that the structure is vertical for all practical purposes but may not be precisely at 90 degrees throughout.

(11) While terms such as “optimal”, “optimize”, “minimal”, “minimize”, “maximal”, “maximize”, and the like may not have certain values associated therewith, if such terms are used herein the intent is that one of ordinary skill in the art would understand such terms to include affecting a value, parameter, metric, and the like in a beneficial direction consistent with the totality of this disclosure. For example, describing a value of something as “minimal” does not require that the value actually be equal to some theoretical minimum (e.g., zero), but should be understood in a practical sense in that a corresponding goal would be to move the value in a beneficial direction toward a theoretical minimum.

Physical Description of an Illustrative Operating Context-Data Storage Systems and Data Centers

(12) There is a commercial demand for high-capacity digital data storage systems, in which multiple data storage devices (DSDs), such as hard disk drives (HDDs), solid-state (e.g., flash memory based) drives (SSDs), tape drives, hybrid drives, and the like are housed in a common enclosure. Data storage systems often include large enclosures that house multiple shelves on which rows of DSDs are mounted. FIG. **2** is a perspective view illustrating one example data storage system, according to embodiments. A data storage system (DSS) **200** may comprise a system enclosure **202** (or “rack **202**”), in which multiple data storage system trays **204** (or otherwise grouped) are housed. Each tray **204** may be placed or slid onto a corresponding shelf or rail within the rack **202**, for example. Each tray is composed of multiple slots, compartments, mounting spaces in which a respective DSD **205** is housed and communicatively coupled with a system controller, such as via a backplane or otherwise. Typically, a rack **202** further houses such a system controller, and may further house switches, storage server(s), application server(s), a power supply, cooling fans, etc. While each tray **204** and constituent DSDs **205** in this example are

illustrated as positioned horizontally within the rack **202**, DSD trays **204** alternatively may be positioned vertically or otherwise within a comparable rack.

(13) Generally, a data center (or, more generally, "mass storage system") may be likened to an extreme version of a data storage system (or multiple data storage systems working together), along with the power, cooling, space, and the like, needed to operate the storage, management, and sharing of data as well as the corresponding network infrastructure (e.g., routers, switches, firewalls, application-delivery controllers, and the like). Expanding on that notion, a "hyperscale" data center generally refers to a facility providing robust, scalable applications and storage services to individuals or other businesses. Exemplary implementations of hyperscale computing include cloud and big data storage, web service and social media platforms, enterprise data centers, and the like, which may consist of thousands of servers linked by an ultra-high speed fiber network. Because businesses depend on the security of their data stored in data centers, the security of a data center itself as well as constituent data storage devices are paramount concerns.

(14) An example data storage system may comprise multiple DSDs such as SSDs and/or HDDs, each communicative with and under the control of a system controller via a communication interface circuitry according to a corresponding communication protocol. FIG. **3** is a block diagram illustrating a data storage system architecture, according to an embodiment. Example architecture **300** illustrates a data storage system **200** that comprises multiple data storage devices (DSDs) **304***a* (DSD1), **304***b* (DSD2), and **304***n* (DSDn), where n represents an arbitrary number of DSDs (e.g., SSDs and/or HDDs) that may vary from implementation to implementation. Each DSD **304***a*-**304***n* is communicative with and under the control of a storage system controller **312**, via a communication interface **322** (e.g., electronic circuitry including electrical connectivity means) according to a corresponding communication protocol **323**. Each DSD **304***a*-**304***n* includes corresponding non-volatile memory (NVM) **306** (e.g., typically in the form of electronic non-volatile memory such as non-volatile memory components **170***a*-**170***n* of FIG. **1**A in the case of SSDs, or spinning magnetic disk media such as recording medium **120** of FIG. **1**B in the case of HDDs) controlled by a respective DSD controller **308** ("DSD CNTLR" **308**). Each DSD controller **308** includes at least a memory **309** and a processor **311**, and the system controller **312** includes at least a memory **313** and a processor **315**.

(15) Processing, functions, procedures, actions, method steps, and the like, that are described herein as being performed or performable by a storage device controller such as DSD controller **308** of each DSD **304***a*-**304***n* (or by system controller **312** of data storage system **200**), may include enactment by execution of one or more sequences of instructions stored in one or more memory units and which, when executed by one or more processors, cause such performance. For example, and according to an embodiment, the DSD controller **308** comprises an application-specific integrated circuit (ASIC) comprising at least one memory unit (e.g., ROM memory **309**) for storing such instructions (such as firmware, for a non-limiting example) and at least one processor (e.g., processor **311**) for executing such instructions, enabling detection of the removal of a storage device from a data storage system such as data storage system **200**. More broadly, DSD controller **308** may be embodied in any form of and/or combination of software, hardware, and firmware. An electronic controller in this context typically includes circuitry such as one or more processors for executing instructions, and may be implemented as a System On a Chip (SoC) electronic circuitry, which may include a memory, a microcontroller, a Digital Signal Processor (DSP), an ASIC, a Field Programmable Gate Array (FPGA), hard-wired logic, analog circuitry and/or a combination thereof, for non-limiting examples. Firmware, i.e., executable logic (e.g., programming code) which may be stored in or read into DSD memory **309**, includes machine-executable instructions for execution by the respective DSD controller **308** in operating each respective DSD **303***a*-**304***n*. Similarly, controller memory **313** includes machine-executable instructions for execution by the controller **312** in operating DSS **200**.

(16) The data storage system **200** may be communicatively coupled with a host **350**, which may be

embodied in a hardware machine on which executable code executes (for non-limiting examples, a computer or hardware server, and the like), or as software instructions executable by one or more processors (for non-limiting examples, a software server such as a database server, application server, media server, and the like). Host **350** generally represents a client of the data storage system **200**, and has the capability to make read and write requests (input/output or "I/O") to the data storage system **200**. Note that the system controller **312** may also be referred to as a "host" because the term is often generally used in reference to any device that makes I/O calls to a data storage device or an array of devices, such as DSDs **304***a*-**304***n*. Host **350** interacts with one or more DSDs **304***a*-**304***n* via the interface **322** (e.g., a physical and electrical I/O interface) for transferring data to and from the DSDs **304***a*-**304***n*, such as via a bus or network such as Ethernet or Wi-Fi or a bus standard such as Serial Advanced Technology Attachment (SATA), PCI (Peripheral Component Interconnect) express (PCIe), Small Computer System Interface (SCSI), or Serial Attached SCSI (SAS), for non-limiting examples.

Context

(17) Recall that physical removal of a storage device from a rack of a data storage system or otherwise, in settings such as a storage server, is highly irregular and may be the outcome of a malicious action. For some applications, it would be critical to avoid data reaching an attacker. However, the controlling code is no longer capable of protecting the storage device and the data stored thereon once it is disconnected and/or removed from the system. In this context, when power to the storage device is dropped there is a challenge to discern between a malicious removal of the storage device and an occasional non-malicious power drop. Thus, detecting specifically the removal of a storage device from a storage rack, and responding according to pre-configured host settings, may be beneficial.

Detecting Removal of a Data Storage Device

(18) FIG. **4** is a diagram illustrating a system for device removal detection, according to an embodiment. The approaches described herein for detecting the removal of a data storage device from a data storage system utilize aspects of the mechanism of connection of the storage device. Removal detection means **400** comprises an electrically-conductive first pin **402**, of a device electrical connector **403** ("device connector **403**") of a storage device **401**, and an electrically-conductive second pin **404** of the device connector **403**, which is longer than the first pin **402**. In the diagram of FIG. **4**(A) both the short ("first") pin **402** and the long ("second") pin **404** are electrically connected with a host electrical connector **405** ("host connector **405**") and thus powered (e.g., and connected to ground (GND) via a pull-down resistor), where the host connector **405** along with the device connector **403** cooperatively correspond to an electrical and physical interface to a host as described elsewhere herein. Removal detection means **400** further comprises a power drop detection circuit **406** of a device controller **408** (see also, e.g., DSD controller **308** of FIG. **3**), configured for measuring the voltage associated with, corresponding to, across each of the pins **402**, **404**. For non-limiting examples, the electrical connector **403** may be physically connected to a host connector **405** constituent to a system backplane or some other form of a communications bus, a system controller board/circuitry, and the like.

(19) Generally, when the storage device **401** is removed (e.g., electrically disconnected from the host and thus the host connector **405**) without notice, the difference in pin length of pin **402** and pin **404** causes the shorter pin **402** to be removed before the longer pin **404**. This temporary change in the supplied power to the electrical connection corresponding to device connector **403** and host connector **405** enables the passing of an indication that this occurrence is a device removal rather than a random power drop. With a random power drop event both pins **402**, **404** likely lose power simultaneously and, therefore, such an event does not and is not intended to qualify as a device removal detection and no subsequent data destruction activity is warranted.

(20) The diagram of FIG. **4**(B) illustrates the storage device **401** in the process of removal, whereby the short pin **402** is removed, disconnected, detached from the host connector **405** before the long

pin **404** is removed, and the power drop detection circuit **406** (“detection circuit **406**”) is configured to detect the short pin **402** removal while the long pin **404** remains electrically connected to the host connector **405**. This situation is considered a potential removal state, in which the short pin **402** has lost power (e.g., pull-down (GND) detected) while the long pin **404** is still powered. According to an embodiment, here the detection circuit **406** sets or causes the setting of a flag indicating the short pin **402** disconnection as detected by short pin **402** power reduction/removal. Furthermore and according to an embodiment, setting such a flag may trigger preparation for a successive step or action, i.e., while there is still power supplied to the storage device **401**. This preparation may be especially pertinent to hard disk drive DSDs which may not be configured or provisioned with much of a source of power, e.g., for data destruction, after main power is inadvertently removed from such a device (e.g., in contrast to an enterprise storage SSD which is likely provisioned with one or more power loss capacitors, such as power loss capacitor **174** of FIG. **1**A), such as in an emergency power off (EPO) scenario.

(21) The diagram of FIG. **4**(C) illustrates the storage device **401** with both pins **402**, **404** removed, disconnected, detached from the host connector **405**, whereby the detection circuit **406** detects the disconnection of both pins. This situation now implies a malicious device removal, in which both the short pin **402** and the long pin **404** have lost power (e.g., pull-down (GND) detected) because the device controller was not notified of this disconnection action in advance. According to an embodiment, here the detection circuit **406** sets or causes the setting of another flag, indicating the long pin **404** disconnection as detected by long pin **404** power reduction/removal.

(22) FIG. **5** is a flow diagram illustrating a method for detecting removal of a storage device, according to an embodiment. Processing, functions, procedures, actions, method steps, and the like, that are described herein may include enactment by execution of one or more sequences of one or more instructions stored in one or more memory units and which, when executed by one or more processors, cause such performance. Therefore, the computing process or procedure of FIG. **5** may be implemented for execution as one or more sequences of one or more instructions stored in one or more memory units and which, when executed by one or more processors, cause performance of the blocks illustrated in FIG. **5**. Referenced controllers may be embodied in any form of and/or combination of software, hardware, and firmware, such as an application-specific integrated circuit (ASIC) comprising at least one memory unit for storing such instructions and at least one processor for executing such instructions. For example, the method of FIG. **5** may be implemented for execution as one or more sequences of one or more instructions, e.g., firmware, stored or read into one or more memory units, e.g., memory **309** of FIG. **3**, and which when executed by one or more processors, e.g., processor(s) **311** of FIG. **3**, to cause performance of the illustrated method steps.

(23) According to an embodiment, the security mode embodied in the method of FIG. **5** is enabled on a storage device (see, e.g. storage device **401** of FIG. **4**) by an enabling command from the host (see, e.g., host **350** of FIG. **3**). Such a command represents an agreement between the host and the storage device to activate the described security mode. Similarly and according to an embodiment, the data destruction portion alone may be enabled on the storage device via a respective command from the host.

(24) At block **502**, determine that a conductive first pin of an electrical connector of a storage device is disconnected from a mating electrical connector. For example, power drop detection circuit **406** (FIG. **4**) of device controller **408** (see also, e.g., DSD controller **308** of FIG. **3**) senses a power/voltage drop or power loss at short pin **402** (FIG. **4**) of device connector **403** of storage device **401** (FIG. **4**). Consistent with the diagram of FIG. **4**(B) the state of storage device **401** is considered in the process of removal, whereby the short pin **402** is removed, disconnected, detached from the host connector **405** before the long pin **404** is removed. As described and according to an embodiment the detection circuit **406** may cause the raising or setting of a flag indicating the short pin **402** is disconnected from the host connector **405**, which may trigger preparation for a successive step or action. For a non-limiting example, storage device **401** may

identify and/or store in local memory a portion of the operating code involving a form of data destruction (e.g., the decryption key, as described in more detail elsewhere herein), such that the key may be readily erased responsive to a complete device removal detection determination.
(25) With reference to the right side of the flow diagram of FIG. **5**, at block **504**, determine at a later time that a conductive second pin of the electrical connector of the storage device is disconnected from the mating electrical connector, wherein the second pin is longer than the first pin. For example, power drop detection circuit **406** senses a power/voltage drop or power loss at long pin **404** (FIG. **4**) of device connector **403** of storage device **401**. Consistent with the diagram of FIG. **4**(C) this situation now implies a malicious device removal, in which both the short pin **402** and the long pin **404** have lost power thereby indicating removal, disconnection, detachment from the host connector **405**. As described and according to an embodiment, here the detection circuit **406** may cause the raising or setting of another flag indicating both the short pin **402** and the long pin **404** are disconnected from the host connector **405**. Otherwise, if power drop detection circuit **406** does not sense a power drop or power loss at long pin **404** subsequent to sensing the power drop or power loss at short pin **402** (block **502**), then device controller **408** determines that a more common non-malicious device power aberration occurred.
(26) At block **506**, responsive to determining that the second pin was disconnected after a predetermined period of time after determining that the first pin was disconnected (at block **504**), determine that the storage device has been disconnected from the mating electrical connector. For example and according to an embodiment, disconnection of and consequent loss or drop in power to both the short pin **402** and the long pin **404** is sufficient to conclude that storage device **401** has been disconnected from the host connector **405** (see also, e.g., host **350** of FIG. **3**), rather than a random power drop to the pins **402**, **404**, of storage device **401**, and therefore that a malicious device removal has occurred. Indication of device removal may be conducted by capturing the voltage drop event to memory, without the need to use additional power.
(27) Thus, at block **508**, responsive to determining that the storage device has been disconnected, the storage device performs a data destruction action corresponding to data stored on the storage device. As discussed, according to an embodiment the data destruction is responsive to previously receiving an enabling command or request from the host. This response may involve some action or operation to prevent the attacker from reading the data in the storage device **401**, and the action should be of low power consumption because the storage device **401** is detached from a power source at this point. In the context of an SSD, one or more “power loss” capacitor(s) on a constituent PCB (printed circuit board) may supply enough energy to perform a data destruction action. For a non-limiting example, in an encrypted storage device **401** the device controller **408** of storage device **401** may erase the cryptographic key(s) associated with or corresponding to the data stored on storage device **401**, e.g., by destroying, deleting, wiping a small amount of data and/or programming code. Here, while the data remains on the storage device **401** itself, by erasing the original encryption key the data is effectively impossible to decrypt and, therefore, the data is rendered unrecoverable. For another non-limiting example, device controller **408** may toggle or set the storage device **401** into a device “lock mode” to render the device inoperable.
(28) According to an embodiment, rather than performing a data destruction action corresponding to the data stored on the storage device **401**, device controller **408** may otherwise trigger a data destruction action corresponding to data stored on the storage device **401**. For a non-limiting example, storage device **401** may store in local memory, such as memory **309** (FIG. **3**), an indication that the storage device **401** has been disconnected. Hence, upon reading such indication upon power being restored to the storage device **401**, the storage device **401** formats/reformats itself. In the case of SSDs and flash memory generally, formatting typically entails a block erase procedure on every storage element in the flash array, e.g., by setting the voltage level on each storage element to a significantly higher level than the standard operating voltage and then dropping it to ground. This data destruction embodiment responsive to a device removal

determination may apply to both encrypted and unencrypted drives, and may also apply to devices without Power Loss Protection (PLP) which may be unable to perform extensive operations after power is down.

(29) With reference to the left side of the flow diagram of FIG. **5**, at block **503***a*, responsive to determining that the first pin was disconnected, determine that the second pin is still connected to the mating electrical connector and, at block **503***b*, start a timer corresponding to a predetermined period of time. For example, after power drop detection circuit **406** senses a power/voltage drop or power loss at short pin **402** of device connector **403** of storage device **401** (at block **502**), power drop detection circuit **406** determines at block **503***a* that the power to long pin **404** indicates that long pin **404** of device connector **403** is still connected to host connector **405**. Responsively, device controller **408** starts a timer corresponding to the predetermined period of time consistent with block **506**. If it is determined that the long pin **404** was disconnected after that predetermined time after detecting disconnection of the short pin **402** (at block **502**) and corresponding to the timer set at block **503***b*, then it is concluded that the storage device **401** has been disconnected from the host connector **405**, in accordance with block **506**.

(30) According to an embodiment, an acceleration meter **410** (FIG. **4**) may be integrated into the device controller **408** to measure the acceleration of the storage device **401** at the time of the short pin **402** and/or long pin **404** removal to verify or confirm the device removal detection. In normal operation, there should be no acceleration measured on the acceleration meter **410**. However, during a suspected device removal, some acceleration would be measured by the acceleration meter **410**. This technique may be combined with the foregoing embodiment or work independently of it. For example, in addition to or alternatively to determining that the long pin **404** was disconnected after that predetermined time after detecting disconnection of the short pin **402**, device controller **408** determines acceleration of the storage device and thereby verifies, confirms, affirms that storage device **401** has been disconnected from host connector **405** and that an unexpected device removal has been detected. The combination of conditions (i.e., pin removal, device acceleration) may differ depending, for example, on the certainty level needed to establish device removal, on the sensitivity of the corresponding sensors, and the like.

(31) Otherwise, at block **505**, responsive to determining that the second pin was disconnected before expiration of the timer, determine that a random power drop occurred. According to an embodiment, a host may be notified that there may be a power problem with the storage device, e.g., that a power drop was detected but with no further action needed or performed. For example, if power drop detection circuit **406** determines that the power to long pin **404** indicates that long pin **404** was disconnected from the host connector **405** before the timer ran out, then device controller **408** may notify host **350** that there may be a problem with powering storage device **401**, i.e., that this sequence of events does not qualify as a device removal detection event.

(32) As data security and protection is a critical aspect in the design of resilient storage devices and corresponding data storage systems, having the capability to detect an unexpected, possibly malicious removal of a storage device from its storage system is beneficial. With monitoring the power to each of multiple pins having different lengths, along with monitoring the relative delay in which such pins experience respective power reductions, suitable logic can rapidly discern between a device disconnection/removal event versus a simple device power aberration. In turn, suitable data destruction action can be taken to render the data stored on the storage device inaccessible to the attacker, thereby protecting the storage device even after the device is removed from the storage system. Such action may be taken immediately and completely by utilizing minimal power stored within the storage device or preliminarily and then completely upon repowering of the device, for examples.

Solid State Drive Configuration

(33) As discussed, embodiments may be used in the context of a data storage system including solid-state drives (SSDs). Thus, FIG. **1**A is a block diagram illustrating an example operating

context with which embodiments of the invention may be implemented. FIG. **1**A illustrates a generic SSD architecture **150**, with an SSD **152** communicatively coupled with a host **154** through a primary communication interface **156**. Embodiments are not limited to a configuration as depicted in FIG. **1**A, rather, embodiments may be implemented with SSD configurations other than that illustrated in FIG. **1**A. For example, embodiments may be implemented to operate in other environments that rely on non-volatile memory storage components for writing and reading of data.

(34) Host **154** broadly represents any type of computing hardware, software, or firmware (or any combination of the foregoing) that makes, among others, data I/O (input/output) requests or calls to one or more memory device. For example, host **154** may be an operating system executing on a computer, a tablet, a mobile phone, or generally any type of computing device that contains or interacts with memory, such as host **350** (FIG. **3**). The primary interface **156** coupling host **154** to SSD **152** may be, for example, a storage system's internal bus or a communication cable or a wireless communication link, or the like.

(35) The example SSD **152** illustrated in FIG. **1**A includes an interface **160**, a controller **162** (e.g., a controller having firmware logic therein), an addressing **164** function block, data buffer cache **166**, and one or more non-volatile memory components **170***a*, **170***b*-**170***n*.

(36) Interface **160** is a point of interaction between components, namely SSD **152** and host **154** in this context, and is applicable at the level of both hardware and software. This enables a component to communicate with other components via an input/output (I/O) system and an associated protocol. A hardware interface is typically described by the mechanical, electrical and logical signals at the interface and the protocol for sequencing them. Some non-limiting examples of common and standard interfaces include SCSI (Small Computer System Interface), SAS (Serial Attached SCSI), and SATA (Serial ATA).

(37) An SSD **152** includes a controller **162**, which incorporates the electronics that bridge the non-volatile memory components (e.g., NAND (NOT-AND) flash) to the host, such as non-volatile memory **170***a*, **170***b*, **170***n* to host **154**. The controller is typically an embedded processor that executes firmware-level code and is an important factor in SSD performance.

(38) Controller **162** interfaces with non-volatile memory **170***a*, **170***b*, **170***n* via an addressing **164** function block. The addressing **164** function operates, for example, to manage mappings between logical block addresses (LBAs) from the host **154** to a corresponding physical block address on the SSD **152**, namely, on the non-volatile memory **170***a*, **170***b*, **170***n* of SSD **152**. Because the non-volatile memory page and the host sectors are different sizes, an SSD has to build and maintain a data structure that enables it to translate between the host writing data to or reading data from a sector, and the physical non-volatile memory page on which that data is actually placed. This table structure or "mapping" may be built and maintained for a session in the SSD's volatile memory **172**, such as DRAM (dynamic random-access memory) or some other local volatile memory component accessible to controller **162** and addressing **164**. Alternatively, the table structure may be maintained more persistently across sessions in the SSD's non-volatile memory such as non-volatile memory **170***a*, **170***b*-**170***n*.

(39) Addressing **164** interacts with data buffer cache **166**, in addition to non-volatile memory **170***a*, **170***b*-**170***n*. Data buffer cache **166** of an SSD **152** typically uses DRAM as a cache, similar to the cache in hard disk drives. Data buffer cache **166** serves as a buffer or staging area for the transmission of data to and from the non-volatile memory components, as well as serves as a cache for speeding up future requests for the cached data. Data buffer cache **166** is typically implemented with volatile memory so the data stored therein is not permanently stored in the cache, i.e., the data is not persistent.

(40) Finally, SSD **152** includes one or more non-volatile memory **170***a*, **170***b*-**170***n* components. For a non-limiting example, the non-volatile memory components **170***a*, **170***b*-**170***n* may be implemented as flash memory (e.g., NAND or NOR flash), or other types of solid-state memory available now or in the future. The non-volatile memory **170***a*, **170***b*-**170***n* components are the

actual memory electronic components on which data is persistently stored. The non-volatile memory **170***a*, **170***b*-**170***n* components of SSD **152** can be considered the analogue to the hard disks in hard-disk drive (HDD) storage devices. According to an embodiment, SSD **152** further comprises a power loss capacitor **174**, which can be used as a source of a relatively small amount of stored power to perform data destruction activity subsequent to device removal detection and consequent loss of externally supplied power, as described elsewhere herein.

(41) Furthermore, references herein to a data storage device may encompass a multi-medium storage device (or "multi-medium device", which may at times be referred to as a "multi-tier device" or "hybrid drive"). A multi-medium storage device refers generally to a storage device having functionality of both a traditional HDD (see, e.g., HDD **100**) combined with an SSD (see, e.g., SSD **150**) using non-volatile memory, such as flash or other solid-state (e.g., integrated circuits) memory, which is electrically erasable and programmable. As operation, management and control of the different types of storage media typically differ, the solid-state portion of a hybrid drive may include its own corresponding controller functionality, which may be integrated into a single controller along with the HDD functionality. A multi-medium storage device may be architected and configured to operate and to utilize the solid-state portion in a number of ways, such as, for non-limiting examples, by using the solid-state memory as cache memory, for storing frequently-accessed data, for storing I/O intensive data, for storing metadata corresponding to payload data (e.g., for assisting with decoding the payload data), and the like. Further, a multi-medium storage device may be architected and configured essentially as two storage devices in a single enclosure, i.e., a traditional HDD and an SSD, with either one or multiple interfaces for host connection.

Hard Disk Drive Configuration

(42) As discussed, embodiments may be used in the context of a data storage system including hard disk drives (HDDs). Thus, in accordance with an embodiment, a plan view illustrating an HDD **100** is shown in FIG. **1**B to illustrate exemplary operating components.

(43) FIG. **1**B illustrates the functional arrangement of components of the HDD **100** including a slider **110***b* that includes a magnetic read-write head **110***a*. Collectively, slider **110***b* and head **110***a* may be referred to as a head slider. The HDD **100** includes at least one head gimbal assembly (HGA) **110** including the head slider, a lead suspension **110***c* attached to the head slider typically via a flexure, and a load beam **110***d* attached to the lead suspension **110***c*. The HDD **100** also includes at least one recording medium **120** rotatably mounted on a spindle **124** and a drive motor (not visible) attached to the spindle **124** for rotating the medium **120**. The read-write head **110***a*, which may also be referred to as a transducer, includes a write element and a read element for respectively writing and reading information stored on the medium **120** of the HDD **100**. The medium **120** or a plurality of disk media may be affixed to the spindle **124** with a disk clamp **128**.

(44) The HDD **100** further includes an arm **132** attached to the HGA **110**, a carriage **134**, a voice coil motor (VCM) that includes an armature **136** including a voice coil **140** attached to the carriage **134** and a stator **144** including a voice-coil magnet (not visible). The armature **136** of the VCM is attached to the carriage **134** and is configured to move the arm **132** and the HGA **110** to access portions of the medium **120**, all collectively mounted on a pivot shaft **148** with an interposed pivot bearing assembly **152**. In the case of an HDD having multiple disks, the carriage **134** may be referred to as an "E-block," or comb, because the carriage is arranged to carry a ganged array of arms that gives it the appearance of a comb.

(45) An assembly comprising a head gimbal assembly (e.g., HGA **110**) including a flexure to which the head slider is coupled, an actuator arm (e.g., arm **132**) and/or load beam to which the flexure is coupled, and an actuator (e.g., the VCM) to which the actuator arm is coupled, may be collectively referred to as a head-stack assembly (HSA). An HSA may, however, include more or fewer components than those described. For example, an HSA may refer to an assembly that further includes electrical interconnection components. Generally, an HSA is the assembly configured to

move the head slider to access portions of the medium **120** for read and write operations.

(46) With further reference to FIG. **1**, electrical signals (e.g., current to the voice coil **140** of the VCM) comprising a write signal to and a read signal from the head **110***a*, are transmitted by a flexible cable assembly (FCA) **156** (or "flex cable"). Interconnection between the flex cable **156** and the head **110***a* may include an arm-electronics (AE) module **160**, which may have an on-board pre-amplifier for the read signal, as well as other read-channel and write-channel electronic components. The AE module **160** may be attached to the carriage **134** as shown. The flex cable **156** may be coupled to an electrical-connector block **164**, which provides electrical communication, in some configurations, through an electrical feed-through provided by an HDD housing **168**. The HDD housing **168** (or "enclosure base" or "baseplate" or simply "base"), in conjunction with an HDD cover, provides a semi-sealed (or hermetically sealed, in some configurations) protective enclosure for the information storage components of the HDD **100**.

(47) Other electronic components, including a disk controller and servo electronics including a digital-signal processor (DSP), provide electrical signals to the drive motor, the voice coil **140** of the VCM and the head **110***a* of the HGA **110**. The electrical signal provided to the drive motor enables the drive motor to spin providing a torque to the spindle **124** which is in turn transmitted to the medium **120** that is affixed to the spindle **124**. As a result, the medium **120** spins in a direction **172**. The spinning medium **120** creates a cushion of air that acts as an air-bearing on which the air-bearing surface (ABS) of the slider **110***b* rides so that the slider **110***b* flies above the surface of the medium **120** without making contact with a thin magnetic-recording layer in which information is recorded. Similarly in an HDD in which a lighter-than-air gas is utilized, such as helium for a non-limiting example, the spinning medium **120** creates a cushion of gas that acts as a gas or fluid bearing on which the slider **110***b* rides.

(48) The electrical signal provided to the voice coil **140** of the VCM enables the head **110***a* of the HGA **110** to access a track **176** on which information is recorded. Thus, the armature **136** of the VCM swings through an arc **180**, which enables the head **110***a* of the HGA **110** to access various tracks on the medium **120**. Information is stored on the medium **120** in a plurality of radially nested tracks arranged in sectors on the medium **120**, such as sector **184**. Correspondingly, each track is composed of a plurality of sectored track portions (or "track sector") such as sectored track portion **188**. Each sectored track portion **188** may include recorded information, and a header containing error correction code information and a servo-burst-signal pattern, such as an ABCD-servo-burst-signal pattern, which is information that identifies the track **176**. In accessing the track **176**, the read element of the head **110***a* of the HGA **110** reads the servo-burst-signal pattern, which provides a position-error-signal (PES) to the servo electronics, which controls the electrical signal provided to the voice coil **140** of the VCM, thereby enabling the head **110***a* to follow the track **176**. Upon finding the track **176** and identifying a particular sectored track portion **188**, the head **110***a* either reads information from the track **176** or writes information to the track **176** depending on instructions received by the disk controller from an external agent, for example, a microprocessor of a computer system.

(49) An HDD's electronic architecture comprises numerous electronic components for performing their respective functions for operation of an HDD, such as a hard disk controller ("HDC"), an interface controller, an arm electronics module, a data channel, a motor driver, a servo processor, buffer memory, etc. Two or more of such components may be combined on a single integrated circuit board referred to as a "system on a chip" ("SOC"). Several, if not all, of such electronic components are typically arranged on a printed circuit board that is coupled to the bottom side of an HDD, such as to HDD housing **168**.

(50) References herein to a hard disk drive, such as HDD **100** illustrated and described in reference to FIG. **1**, may encompass an information storage device that is at times referred to as a "hybrid drive". A hybrid drive refers generally to a storage device having functionality of both a traditional HDD (see, e.g., HDD **100**) combined with solid-state storage device (SSD) using non-volatile

memory, such as flash or other solid-state (e.g., integrated circuits) memory, which is electrically erasable and programmable. As operation, management and control of the different types of storage media typically differ, the solid-state portion of a hybrid drive may include its own corresponding controller functionality, which may be integrated into a single controller along with the HDD functionality. A hybrid drive may be architected and configured to operate and to utilize the solid-state portion in a number of ways, such as, for non-limiting examples, by using the solid-state memory as cache memory, for storing frequently-accessed data, for storing I/O intensive data, and the like. Further, a hybrid drive may be architected and configured essentially as two storage devices in a single enclosure, i.e., a traditional HDD and an SSD, with either one or multiple interfaces for host connection.

EXTENSIONS AND ALTERNATIVES

(51) In the foregoing description, embodiments of the invention have been described with reference to numerous specific details that may vary from implementation to implementation. Therefore, various modifications and changes may be made thereto without departing from the broader spirit and scope of the embodiments. Thus, the sole and exclusive indicator of what is the invention, and is intended by the applicant(s) to be the invention, is the set of claims that issue from this application, in the specific form in which such claims issue, including any subsequent correction. Any definitions expressly set forth herein for terms contained in such claims shall govern the meaning of such terms as used in the claims. Hence, no limitation, element, property, feature, advantage or attribute that is not expressly recited in a claim should limit the scope of such claim in any way. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense.

(52) In addition, in this description certain process steps may be set forth in a particular order, and alphabetic and alphanumeric labels may be used to identify certain steps. Unless specifically stated in the description, embodiments are not necessarily limited to any particular order of carrying out such steps. In particular, the labels are used merely for convenient identification of steps, and are not intended to specify or require a particular order of carrying out such steps.

Claims

1. A method for detecting removal of a data storage device from a system, the method comprising: determining that an electrically conductive first pin of an electrical connector of a storage device is disconnected from a mating electrical connector by circuitry detecting a voltage drop corresponding to the first pin; determining at a later time that an electrically conductive second pin of the electrical connector of the storage device is disconnected from the mating electrical connector by circuitry detecting a voltage drop corresponding to the second pin, wherein the second pin is longer than the first pin; responsive to determining that the second pin was disconnected after a predetermined period of time after determining that the first pin was disconnected, determining that the storage device has been disconnected from the mating electrical connector; and responsive to determining that the storage device has been disconnected, triggering a data destruction action corresponding to data stored on the storage device.
2. The method of claim 1, further comprising: responsive to determining that the storage device has been disconnected, the storage device performing a data destruction action corresponding to data stored on the storage device.
3. The method of claim 2, wherein performing a data destruction action includes performing a data destruction action responsive to determining that a host enabled the data destruction on the storage device.
4. The method of claim 2, wherein the storage device performing the data destruction action includes the storage device erasing from memory a cryptographic key corresponding to the data.
5. The method of claim 2, wherein the storage device performing the data destruction action

includes the storage device storing in memory an indication that the storage device has been disconnected, the method further comprising: responsive to reading the indication upon power being restored to the storage device, the storage device formatting itself.
6. The method of claim 1, further comprising: responsive to determining that the first pin was disconnected, determining that the second pin is still connected to the mating electrical connector; and starting a timer corresponding to the predetermined period of time.
7. The method of claim 1, further comprising: prior to triggering the data destruction action, determining acceleration of the storage device, and responsive to determining acceleration of the storage device, determining verification that the storage device has been disconnected; and responsive to determining verification that the storage device has been disconnected, triggering the data destruction action corresponding to data stored on the storage device.
8. The method of claim 7, wherein determining acceleration includes determining acceleration responsive to determining that the second pin is disconnected.
9. An electronic controller for a data storage device (DSD), the controller comprising: means for executing machine-executable instructions; and means for storing one or more sequences of machine-executable instructions which, when executed by the means for executing, cause performance of: determining that a conductive first pin of an electrical connector of the DSD is disconnected from a mating electrical connector by circuitry detecting a voltage drop corresponding to the first pin; determining at a later time that a conductive second pin of the electrical connector of the DSD is disconnected from the mating electrical connector by circuitry detecting a voltage drop corresponding to the second pin, wherein the second pin is longer than the first pin; responsive to determining that the second pin was disconnected after a predetermined period of time after determining that the first pin was disconnected, determining that the DSD has been disconnected from the mating electrical connector; and responsive to determining that the DSD has been disconnected, triggering a data destruction action corresponding to data stored on the DSD.
10. The electronic controller of claim 9, wherein the one or more sequences of machine-executable instructions, when executed by the means for executing, cause further performance of: responsive to determining that the DSD has been disconnected, performing a data destruction action corresponding to data stored on the DSD.
11. The electronic controller of claim 10, wherein performing a data destruction action includes performing a data destruction action responsive to determining that a host enabled the data destruction on the DSD.
12. The electronic controller of claim 10, wherein performing the data destruction action includes erasing from memory a cryptographic key corresponding to the data.
13. The electronic controller of claim 10, wherein performing the data destruction action includes storing in memory an indication that the DSD has been disconnected, wherein the one or more sequences of machine-executable instructions, when executed by the means for executing, cause further performance of: responsive to reading the indication upon power being restored to the DSD, formatting the DSD.
14. The electronic controller of claim 10, wherein: the DSD is a solid-state drive (SSD); and performing the data destruction action includes using power from an internal capacitor to perform the data destruction action.
15. The electronic controller of claim 9, wherein the one or more sequences of machine-executable instructions, when executed by the means for executing, cause further performance of: responsive to determining that the first pin was disconnected, determining that the second pin is still connected to the mating electrical connector; and starting a timer corresponding to the predetermined period of time.
16. The electronic controller of claim 9, wherein the one or more sequences of machine-executable instructions, when executed by the means for executing, cause further performance of: prior to

triggering the data destruction action, detecting an acceleration of the DSD using an internal acceleration meter, and confirming that the DSD has been disconnected based on determining the acceleration of the DSD; and responsive to confirming that the DSD has been disconnected, triggering the data destruction action corresponding to data stored on the DSD.

17. A data storage device (DSD) comprising: non-volatile memory; and an electronic controller comprising: one or more processors configured for executing machine-executable instructions, and one or more memory configured for storing one or more sequences of machine-executable instructions which, when executed by the one or more processors, individually or in combination, cause performance of: determining that a conductive first pin of an electrical connector of the DSD is disconnected from a mating electrical connector by detecting a voltage drop corresponding to the first pin; determining at a later time that a conductive second pin of the electrical connector of the DSD is disconnected from the mating electrical connector by detecting a voltage drop corresponding to the second pin, wherein the second pin is longer than the first pin; responsive to determining that the second pin was disconnected after a predetermined period of time after determining that the first pin was disconnected, determining that the DSD has been disconnected from the mating electrical connector; and responsive to determining that the DSD has been disconnected, triggering a data destruction action corresponding to data stored on the DSD.

18. The data storage device of claim 17, wherein the one or more sequences of machine-executable instructions, when executed by the one or more processors, individually or in combination, cause further performance of: responsive to determining that the DSD has been disconnected, the DSD performing a data destruction action corresponding to data stored on the DSD.

19. The data storage device of claim 18, wherein performing the data destruction action includes storing in memory an indication that the DSD has been disconnected, and wherein the one or more sequences of machine-executable instructions, when executed by the one or more processors, individually or in combination, cause further performance of: responsive to reading the indication upon power being restored to the DSD, formatting the DSD.

20. The data storage device of claim 17, wherein the one or more sequences of machine-executable instructions, when executed by the one or more processors, individually or in combination, cause further performance of: prior to triggering the data destruction action, determining acceleration of the DSD, and responsive to determining acceleration of the DSD, determining verification that the DSD has been disconnected; and responsive to determining verification that the DSD has been disconnected, triggering the data destruction action corresponding to data stored on the DSD.